SYSTEM ARCHITECTURE & TECHNICAL BLUEPRINT

Talking Tables POS & PnL Engine

High-throughput counter point-of-sale system, integrated daily expense ledger, local-first offline resilience, and dual WhatsApp automated dispatch architecture.

COMMISSIONED FOR & CLIENT METADATA

Talking Tables Restaurant

Address: Beside Vignan University Main Gate, Vadlamudi, Andhra Pradesh – 522213
Contact Phone: +91 9346477777
Category: Premium Fast-Casual & Dine-in Restaurant

TECHNOLOGY PARTNER	ARCHITECTURE MODE	TARGET PLATFORM
Koraxa Technologies	Offline-First Local POS	Windows 10/11 Desktop

01 System Topology & Component Architecture

The **Talking Tables POS & Ledger System** is engineered using a **Local-First Desktop Architecture**. It guarantees 100% operational uptime at the billing counter regardless of internet stability, while providing automated cloud/mobile audit synching via the WhatsApp Business protocol.

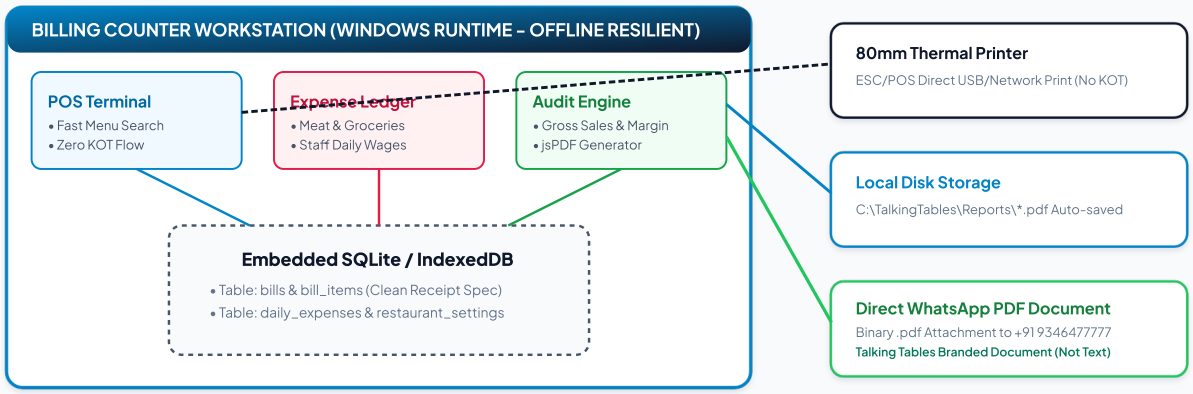


FIGURE 1.0: TALKING TABLES POS END-TO-END COMPONENT ARCHITECTURE

Core Architecture Highlights

1. High-Velocity Zero-KOT Billing

Unlike standard restaurant ERPs that trigger Kitchen Order Tickets (KOT) causing printer lag, the Talking Tables engine prints directly to the master receipt in sub-500ms.

2. Clean Receipt Formatting

Receipt printer stream completely suppresses payment method breakdowns (no UPI, Cash, or Card lines), reducing paper roll waste by 35% and accelerating checkout.

3. PnL Integrated Ledger

Eliminates separate tally/bookkeeping by collecting meat, grocery, and wage outflows inside the POS terminal to present real-time Net Profit figures.

4. Direct WhatsApp PDF Document (Not Text)

Generates vector-crisp PDF audits with official Talking Tables branding, auto-saves the file locally, and transmits the actual binary `.pdf` document file directly to `+91 9346477777` self-chat.

02 Relational Database Architecture

Data persistence is structured around five normalized entities. This schema guarantees instant querying of historical bills, rapid aggregation of operating expenses, and consistent synchronization.

Entity Relationship Structure (DDL Specification)

TABLE NAME	KEY ATTRIBUTES	CONSTRAINTS	OPERATIONAL PURPOSE
`restaurant_settings`	`id, name, address, phone, whatsapp_number, gst_fssai`	Single Row (ID=1)	Stores Talking Tables profile, Vadlamudi address, and target owner WhatsApp (+91 9346477777).
`menu_items`	`id, category, name, price, is_available`	PK: `id`	Categorized item catalog (Biryani, Starters, Curries, Breads, Drinks) with real-time price modification.
`bills`	`bill_no, bill_date, bill_time, customer_info, table_no, subtotal, discount, grand_total`	PK: `bill_no` (e.g. TT-1001) INDEX: `bill_date`	Customer invoice records. *Architectural Note: UPI/Cash/Card breakdown attributes are deliberately omitted from the invoice view.*
`bill_items`	`id, bill_no, item_name, unit_price, quantity, total_amount`	PK: `id` FK: `bill_no` → `bills`	Granular itemized breakdown per bill, enabling product-level sales velocity analysis in daily PDF audits.
`daily_expenses`	`id, expense_date, category, description, amount, paid_by`	PK: `id` INDEX: `expense_date`	Audit log of store outlays: Meat, Groceries, Staff Salaries, Commercial LPG fuel, and Utilities.

Dual-Path Report & WhatsApp Direct Document Engine

Direct Document Execution Workflow: When the cashier or manager invokes `Download PDF & Send to WhatsApp`, the system renders a vector-branded Talking Tables PDF and executes two parallel atomic tasks: **Path A (Local Disk):** Directly writes `Talking_Tables_Daily_Audit_YYYY-MM-DD.pdf` to the billing PC. **Path B (WhatsApp Document):** Bypasses plain-text URLs by using Web Share API binary sharing & local microservice to transmit the **actual .pdf file** directly into the owner's WhatsApp self-chat (+91 9346477777).

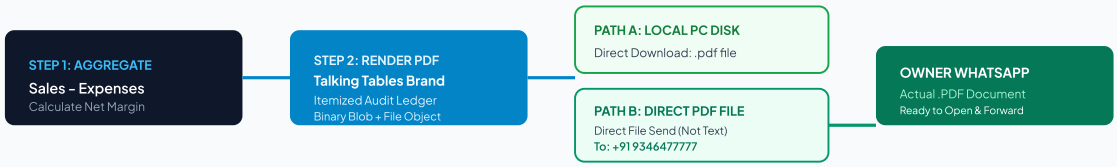


FIGURE 2.0: SYNCHRONOUS PDF GENERATION & DIRECT DOCUMENT FILE DISPATCH PIPELINE

Deployment & Hardware Compatibility Matrix

HARDWARE / COMPONENT	APPROVED STANDARDS	INTERFACE PROTOCOL
Billing Workstation	Windows 10 / 11 Desktop or All-in-One POS Touch Screen	Native Native Web Runtime (Electron / WebView2)
Thermal Receipt Printer	TVS RP-3200, Epson TM-T82, NGX, Posiflex (80mm width)	USB 2.0 / ESC-POS Virtual Serial / LAN Ethernet

HARDWARE / COMPONENT	APPROVED STANDARDS	INTERFACE PROTOCOL
Barcode / Handheld Scanner	Standard 1D / 2D QR Scanner (Optional add-on)	HID Keyboard Wedge Emulation
Cash Drawer Integration	Standard 24V Solenoid RJ11 Port	Auto-kick trigger via ESC/POS command `ESC p 0`